

Feature-matched masking reveals a dissociation between iconic and working memory

Abstract

Visual working memory preserves perceptual information across delay, yet the representational basis of that preservation remains contested. Influential sensory recruitment accounts propose that retention depends on sustained or reinstated perceptual codes in early visual cortex. Under some proposals, this predicts that interference delivered to the retinotopic location of a remembered item during delay should disrupt memory-guided report. In three experiments, we applied a dynamic, masking manipulation across two retention regimes, a 400-ms iconic interval and a multi-second working memory delay, to explore whether and how sensory input interacts with retained visual information.

The results revealed a striking dissociation: consistent with disruption of a fragile, sensory-format trace, masks delivered during a 400ms iconic window more than tripled guess rates while leaving precision of response intact (Experiment 1). In contrast, the identical masks left working-memory performance unchanged at every load, even as load itself reliably impaired performance (Experiment 2). A within-subjects manipulation of mask onset reproduced the dissociation: only masks coincident with sample offset disrupted report, while mask onsets 100ms or more later left performance intact (Experiment 3).

These results establish a critical constraint on sensory recruitment accounts: once consolidation is complete, multi-second working memory retention does not depend on the sensory-format code that competes with ongoing input for the same feature channels.

Keywords: visual working memory, iconic memory, sensory recruitment, masking, representational format, psychophysics

1. Introduction

Visual working memory supports short-term retention of visual information for comparison, decision, and action. We can compare displays across seconds, maintain target features during search, and recover remembered features while the perceptual system continues to process ongoing input. Any account of visual working memory must therefore explain how perceptual information remains available after stimulus offset during ongoing sensory processing.

1.1. Sensory recruitment and the question of representational format

One influential account depends on sensory recruitment. On this view, remembered content is maintained through sustained perceptual representations in early visual cortex ([Harrison & Tong, 2009](#); [Serences et al., 2009](#); [Serences, 2016](#); [Sreenivasan, Curtis, & D'Esposito, 2014](#)). In nonhuman primates, delay-period activity appears in sensory and sensory-adjacent regions including V4, IT, and MT ([Miller et al., 1993](#); [Hayden & Gallant, 2013](#); [Chelazzi et al., 1998](#); [Pasternak & Greenlee, 2005](#); [Bisley et al., 2001](#); [2004](#); [Zaksas et al., 2001](#)). In humans, remembered features can be decoded from early visual cortex during a delay, and the fidelity of those patterns covaries with psychophysical measures of mnemonic precision ([Harrison & Tong, 2009](#); [Serences et al., 2009](#); [Xing et al., 2013](#); [Emrich et al., 2013](#); [Ester et al., 2013](#)). That feature-specific representations in early sensory areas track the content and quality of what is being remembered is precisely why sensory recruitment is compelling: the same code that supports perception appears to carry mnemonic information forward.

In its most basic formulation, sensory recruitment treats working memory maintenance as continued or reinstated perceptual encoding: the system bridges the retention interval by keeping the original sensory code active in the same feature-selective, retinotopically organized populations that encode it ([Serences, 2016](#); [Sreenivasan, Curtis, & D'Esposito, 2014](#)). Such channel-specificity also carries a direct empirical prediction: if remembered orientations are maintained in the same sensory populations that support encoding, then subsequent input that drives those same populations during a delay should perturb the mnemonic signal.

The primary evidence for sensory recruitment comes from decoding experiments, which are consistent with the theory, but not dispositive. That feature-specific delay-period activity is decodable in early visual cortex ([Harrison & Tong, 2009](#); [Emrich et al., 2013](#); [Ester et al., 2013](#)) does not by itself establish that the region's activity constitutes the substrate on which report depends ([Riggall & Postle, 2012](#); [Xu, 2017](#)). A region may carry feature information as a byproduct of feedback from higher areas (e.g., [van Kerkoerle et al., 2017](#); [Lawrence et al., 2018](#)) as a readout-preparation signal that anticipates use rather than constituting the basis for storage (e.g., [Stokes, 2015](#); [Wolff et al., 2017](#); [Myers, Stokes, & Nobre 2017](#)), or as an attentional or priority trace coupled to an item's location rather than a content-specific mnemonic code (e.g., [Bettencourt & Xu, 2016](#); [Sprague & Serences 2013](#))- each of which would be recoverable by decoding without being causally necessary for retention.

This distinction between presence and dependence constitutes a central tension in the field, and these dissociations between neural decodability and behavioral performance

have motivated accounts in which working memory does not maintain a simple copy of the sensory code. One prevalent model suggests that representations may be transformed within visual cortex into orthogonalized formats organized around task demands rather than the retinotopic and feature-specific structure of the original sensory encoding, possibly via rotation, compression, or categorical recoding ([Murray et al., 2017](#); [Kiyonaga & Serences, 2025](#)). They may coexist with ongoing sensory encoding within overlapping neural populations but occupy separable subspaces that insulate mnemonic from sensory content ([Rademaker, Chunharas, & Serences, 2019](#)).

Other accounts place maintained representations outside the feedforward sensory pathway altogether: storage may enlist top-down feedback signals (e.g., from fronto-parietal regions) to early visual cortex that target feedback-recipient layers rather than feedforward-recipient layers ([van Kerkoerle et al., 2017](#); [Lawrence et al., 2018](#)) or in activity-silent synaptic states that carry no sustained firing and are therefore invisible to conventional decoding ([Stokes, 2015](#); [Wolff et al., 2017](#)).

When fully fleshed out, some of these theories may predict that ongoing feedforward input should disrupt working memory maintenance and subsequent performance. For example, a representation orthogonalized within V1 but still anchored to sensory populations via feedforward pathways would remain susceptible to interference targeting that pathway; a representation maintained by feedback signals or by activity-silent synaptic states would not. Knowing how and when incoming sensory signals (that are not task relevant) disrupt working memory should therefore place constraints on the viability of specific recruitment-oriented theories. Accordingly, we sought to test whether maintenance *depends* on a sensory representation that remains coupled to the same feature channels that support initial encoding. What is central for the current investigation is whether a durable store of the visual stimulus remains available even after disruption of iconic memory and the sensory representations that subserve the feedforward sweep.

1.2. The Iconic store and sensory-format dependence

The question of sensory-format dependence already has a clear answer in one retention regime. A rich literature characterizes *iconic memory* as a brief, high-capacity sensory store that retains the retinotopic organization and low-level feature selectivity of the original sensory encoding ([Bradley & Pearson, 2012](#); [Coltheart, 1980](#); [Sperling, 1960](#)). We expect that iconic representations remain embedded in the same sensory machinery that processes incoming input: as such, they remain vulnerable to overwrite from subsequent stimulation that recruits overlapping feature channels ([Averbach & Coriell, 1961](#); [Breitmeyer & Öğmen, 2006](#); [Di Lollo, 1980](#)). The vulnerability of this sensory substrate has been characterized extensively in visual masking paradigms, with especially strong disruption when masks share feature content and retinotopic position

with targets ([Averbach & Coriell, 1961](#); [Breitmeyer & Ögmen, 2006](#); [Di Lollo, 1980](#)). Of course, iconic memory and visual working memory are not equivalent: they differ in capacity, time-course, and putative locus ([Sperling 1960](#), [Phillips, 1974](#)), but the most straightforward sensory recruitment accounts commit them to a shared representational substrate ([Harrison & Tong, 2009](#); [Serences et al., 2009](#); [Ester et al., 2013](#)). That is, to the extent that residual activity in V1 is dependent on the feedforward signal for maintenance, the supported representations will be vulnerable to masking.

Feature-matched masks therefore engage precisely the substrate that sensory recruitment accounts identify as the working memory store. This makes iconic memory a uniquely useful positive control for sensory-format dependence. To the extent that any storage mechanism depends on the rapid feedforward coding of early visual features—whether via an immediate sensory representation, sustained attention to those features, or rapid recoding within early visual cortex—it may inherit iconic vulnerability to feature-matched interference. A null feature-matched masking effect on working memory under conditions that successfully disrupt iconic memory would then implicate a representational substrate insulated from bottom-up sensory drive, protected from the very interference that overwrites the iconic store.

Existing evidence has yet to resolve this prediction cleanly. Indeed, a variety of masking studies report working memory impairment under sensory interference ([Phillips, 1974](#); [Sreenivasan & Jha, 2007](#); [Vogel et al., 2006](#); [Xu, 2017](#)), but at least three limitations complicate interpretation. First, interference typically occurred only during early consolidation; once consolidated, working memory was unaffected by masking, leaving open whether maintenance-period representations are truly robust or whether the timing of interference simply missed the vulnerable window. Second, in the few cases where null interference effects on working memory were reported, no sensory baseline was established to verify mask effectiveness, making it impossible to distinguish genuine working memory robustness from an ineffective manipulation ([Sreenivasan & Jha, 2007](#)). Third, masking strength depends critically on target-mask feature correspondence ([Rademaker et al., 2015](#)): masks that share spatial frequency, orientation bandwidth, and retinotopic position with remembered items produce stronger disruption precisely because they recruit overlapping sensory channels ([Bruchmann et al., 2010](#); [Drewes et al., 2023](#); [White & Lorber, 1976](#)). Without establishing that a masking manipulation successfully engages sensory-format representations under matched conditions, a null working-memory outcome carries no clear interpretation.

1.3. The present experiments

To ask whether working memory depends on the same sensory-format representation as iconic memory, we applied the same dynamic, feature-matched masking manipulation across two retention regimes. Experiment 1 characterized the effect of backward masking on retention across a 400-ms iconic interval. Experiment 2 applied the identical manipulation across a five-second working memory delay, crossing mask condition with memory load (1–3 items). Experiment 3 then reproduced this dissociation within a single iconic-scale paradigm, varying only mask onset.

If working memory storage relies on the same sensory-format trace as iconic memory, masks that disrupt iconic availability should produce the same vulnerability signature during the working memory delay. However, masking selectively disrupted iconic memory while leaving working memory performance unchanged. The transition from sensory persistence to working memory maintenance therefore requires a change in representational format to an insulated, durable store.

2. General methods

2.1 Participants

Each experiment recruited 40 college-age students from the Johns Hopkins University SONA Subject Pool to participate in exchange for course credit.

2.2 Apparatus and setup

Stimuli were generated and presented using custom PsychoPy code on a BenQ Zowie desktop computer projecting to a 1920×1080px display with a 100Hz refresh rate. Centrally-fixating observers (enforced via desktop-mounted Eyelink 1000 eyetracker; [SR Research](#)) were tested in a dimly lit, sound-attenuated room, maintained with a forehead and chinrest.

2.3 Stimuli

Stimuli were full-contrast sine-wave gratings presented at fixed isoeccentric (5.14°) locations around a central fixation cross. Sample gratings had a radius of 1.285° , phase of 0 and spatial frequency of 5 cycles per stimulus (scaled with stimulus size in PsychoPy). Mask gratings had a radius of 1.928° and matched the sample gratings in spatial frequency and contrast. On mask trials, target locations were flooded with rapidly changing, high-contrast gratings that shared spatial frequency, contrast, and retinotopic position with the sample. Mask orientations were sampled uniformly with the constraint that successive orientations differed by at least 30° .

Experiment 1 (Iconic)

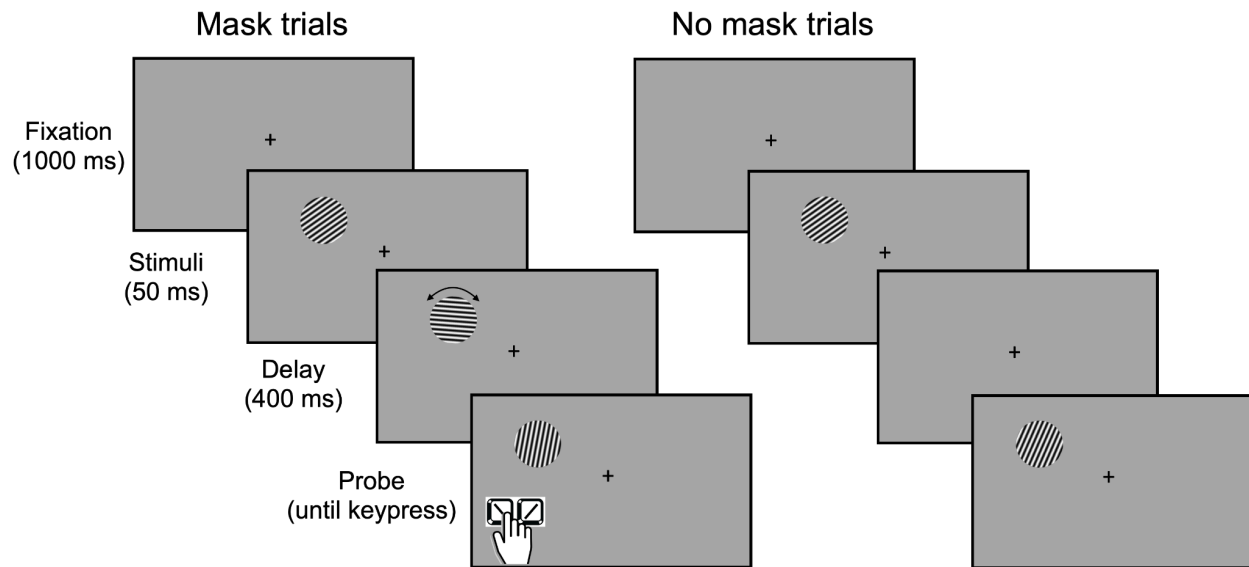


Figure 1. Sample task and trial sequence from Experiment 1 (Iconic task). Each trial began with central fixation. A single peripheral grating appeared briefly (50ms) at one of the predefined stimulus locations, followed by a 400ms retention interval. During the interval, the trial was either unmasked (blank screen) or masked, in which the previously occupied location was filled with rapidly changing, feature-matched mask gratings (orientation updated every 50ms). At test, a single grating appeared at the same location, randomly rotated relative to the sample by $\pm 10^\circ$, $\pm 20^\circ$, $\pm 30^\circ$, or $\pm 40^\circ$. Observers reported whether the orientation change was clockwise or counterclockwise.

3. Experiment 1

Experiment 1 tested whether dynamic, feature-matched masks impair performance in a purely iconic memory task. If performance in this regime depends on a sensory-format trace, then masks that recruit the same orientation channels at the same spatial positions should disrupt access to that trace.

3.1 Experiment 1 Methods

On each trial, a 50ms sample grating appeared at one of six peripheral locations, followed by a 400ms retention interval that was either blank or flooded with feature-matched masks at the sample location (orientation updated every 50ms; **Figure 1**). These timings were chosen to optimize performance with respect to the time-course of iconic memory (e.g., [Coltheart, 1983](#); [Irwin and Yeomans, 1986](#)). At test, a probe appeared in the same location, rotated relative to the sample by $\pm 10^\circ$, $\pm 20^\circ$, $\pm 30^\circ$, or $\pm 40^\circ$. Participants indicated whether the change was clockwise or counterclockwise.

relative to the sample. Each participant completed 80 trials. Mask condition (mask vs. no mask) was manipulated within subjects.

3.2 Experiment 1 Results

Performance was modeled for each participant using a cumulative normal psychometric function (psignifit 4; Schütt et al., 2016), fit separately for mask and no-mask conditions. The psychometric function relates the signed orientation difference between test and sample to the probability of a "clockwise" response (**Figure 2A**). From each fit, we extracted two parameters: precision ($1/\sigma$), indexing the fidelity of the orientation representation, and guess rate (γ), indexing the lower and upper asymptote, corresponding to the probability that orientation information was entirely unavailable at test. Twelve participants were excluded due to at least one invalid psychometric fit (**Supplementary S1**), leaving $n = 28$ in subsequent analyses.

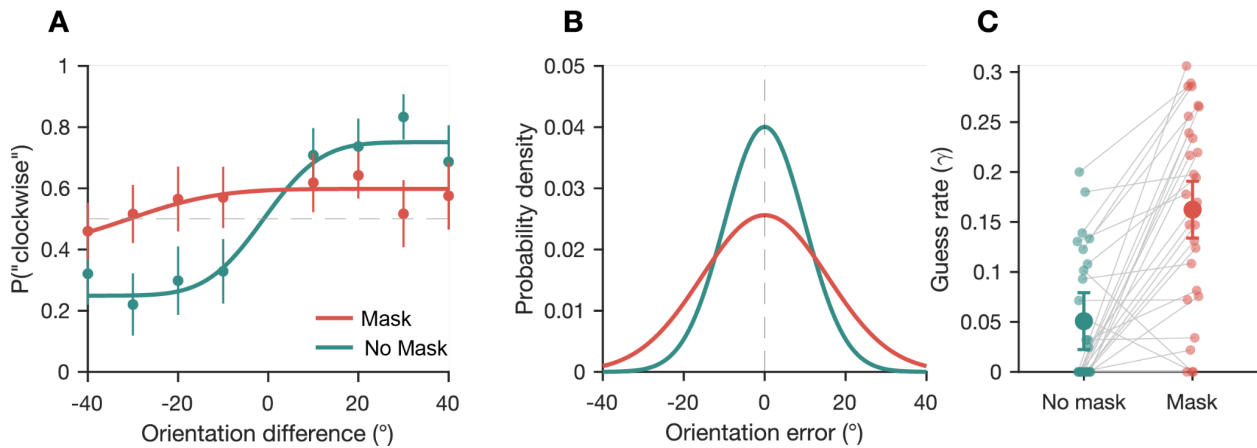


Figure 2: Masking selectively increases guess rate during iconic retention. (A)

Group psychometric functions for the masked (orange) and unmasked (teal) conditions in Experiment 1. Points show the mean proportion of "clockwise" responses across participants at each signed orientation difference (test minus sample, degrees); solid curves are cumulative Gaussian fits to the aggregated binomial counts across subjects. Error bars denote 95% between-subject confidence intervals. **(B)** Precision depicted as Gaussian probability density functions depicting the internal noise distribution implied by fitted sigmas from group-level psychometric fits in **(A)**. **(C)** Guess rate (γ) from per-subject psychometric fits. Individual participant data (small dots); group means with within-subject 95% Cousineau-Morey confidence intervals (large dots with error bars).

3.2.1 Precision

Precision estimates in the iconic task did not differ reliably between mask and no-mask trials ($t(27) = 1.763$, $p > .05$, $BF_{10} = 0.87$; **Figure 2B**). Thus, when orientation information was successfully retained, its fidelity was not degraded by masking.

3.2.2 Guess rate

In contrast, masking robustly increased guess rates. The γ parameter was more than three times higher in mask trials ($M = 0.162$) compared to no-mask trials ($M = 0.051$), $t(27) = 5.686$, $p < .001$, $BF_{10} = 11575$. This masking effect was expressed as a reduction in the asymptote of the psychometric function, consistent with a total loss of usable sensory information, rather than degraded precision (**Figure 2C**).

3.2.3 Effective precision

Because masking substantially elevates guess rates, raw precision estimates ($1/\sigma$) may be inflated through a known slope-asymptote trade-off in cumulative Gaussian models (Wichmann & Hill, 2001; Prins, 2012). We therefore computed an effective precision measure that scales $1/\sigma$ by the proportion of trials containing usable information (**Supplementary S2**). Effective precision did not differ between conditions ($t(27) = 0.775$, $p = .445$, $BF_{10} = 0.26$), confirming that masking reduces iconic availability without altering the fidelity of the surviving iconic representation.

Experiment 1 establishes the all-or-none behavioral signature of feature-matched interference in a sensory-dependent retention regime. Masking did not measurably degrade the fidelity of remembered items; instead, it increased the probability that usable information was lost altogether.

4. Experiment 2 (VWM)

Experiment 2 applied the identical masking manipulation across a multi-second delay to ask whether the same feature-matched interference that eliminated iconic availability would similarly impair working memory performance. The paradigm and stimuli matched Experiment 1, extended across a multi-second delay (5s) and longer encoding duration (1.2s) consistent with the known time-course of visual working memory. We additionally crossed mask condition with memory load (1-3 items) to test a specific prediction of sensory recruitment: if maintenance occupies the same feature-selective populations that encoded the samples, then feature-matched interference should scale with load (e.g., Emrich et al., 2013).

4.1 Experiment 2 Methods

The spatial layout, stimulus properties, and mask characteristics remained identical to Experiment 1. Mask condition (mask vs. no mask) and memory load (1-3 items) were manipulated within subjects. 24 probe detection trials were randomly interspersed to ensure sustained attention during the delay.

4.2 Experiment 2 Results

The same psychometric modeling procedure was applied at each level of load (1-3 items) and mask condition (**Figure 3A**). Eleven participants were excluded due to at least one invalid psychometric fit across the six conditions (3 loads X 2 mask conditions), leaving $N = 29$.

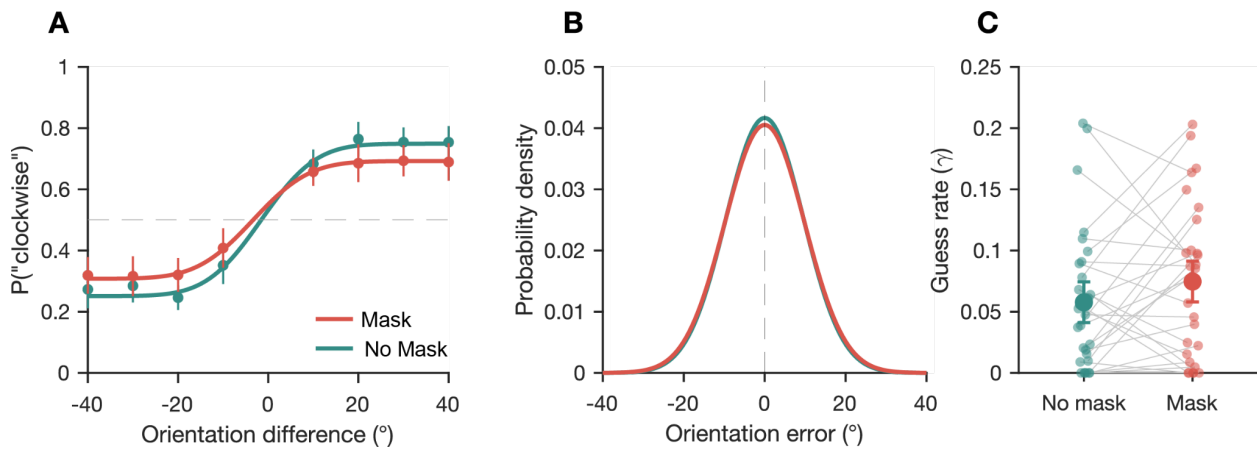


Figure 3. Delay-period masking leaves working memory performance intact. (A)

Group psychometric functions (collapsed across load) for the masked (orange) and unmasked (teal) conditions in Experiment 2. **(B)** Precision depicted as Gaussian probability density functions. Curves depict the internal noise distribution implied by fitted sigmas from group-level psychometric fits in **(A)**. **(C)** Guess rate (γ). Individual participant data (small dots); group means with within-subject 95% Cousineau-Morey confidence intervals (large dots with error bars).

4.2.1 Precision

A 2 (mask) X 3 (load) repeated-measures ANOVA on precision estimates revealed no significant main effect of load, $F(2, 56) = 0.794$, $p > .1$, no main effect of mask, $F(1, 28) = 1.409$, $p > .1$, and no load X mask interaction, $F(2, 56) = 0.181$, $p > .5$ (**Figure 4A**). Bayesian comparisons converged (BF_{10} for mask = 0.133, BF_{10} for load = 0.014, BF_{10} for interaction = 0.007), providing substantial-to-strong evidence favoring the null. Mean precision declined numerically with increasing load (Load 1: $M = 0.274$; Load 2: $M = 0.214$; Load 3: $M = 0.176$), but this trend was not statistically reliable. Collapsed across

load, mask and no-mask trials did not differ in precision, $t(28) = -1.187$, $p = .245$, $BF_{10} = 0.38$ (**Figure 3B**).

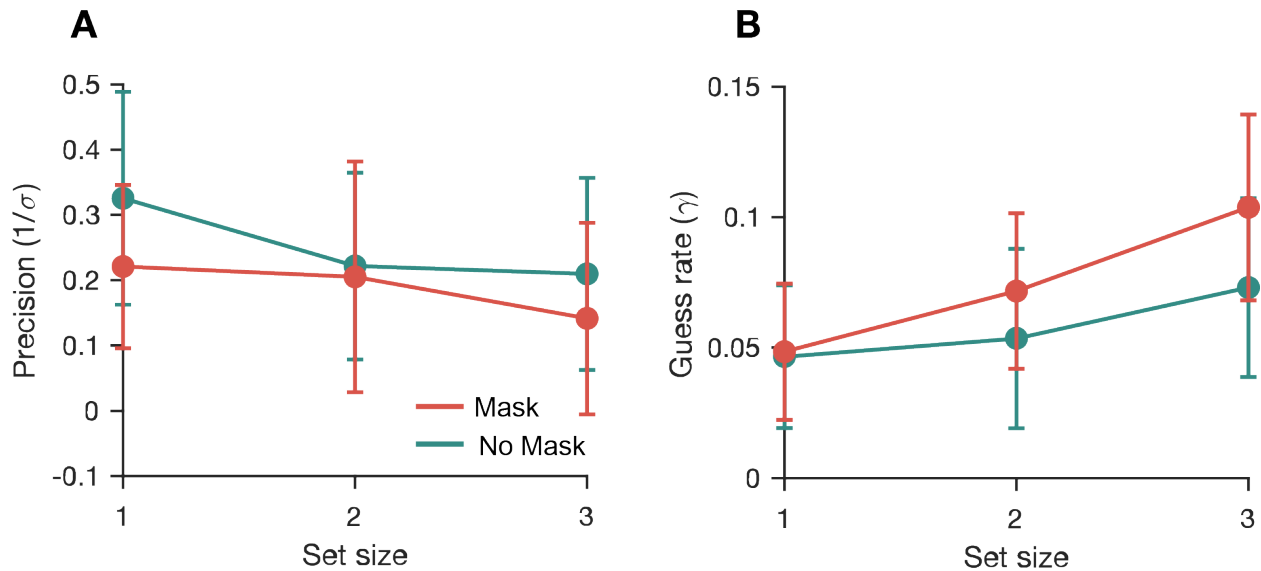


Figure 4. Load-dependent degradation without masking effects in visual working memory. (A) Model-derived precision ($1/\sigma$) estimates as a function of item load (1, 2, 3), shown separately for the no-mask (teal) and mask (orange) condition. Precision declines systematically with increasing load, and does so independently of masking condition. There was no main effect of masking. **(B)** Model-derived guess rate (γ) estimates across load, plotted separately for no-mask and mask trials. Guessing increases with load, consistent with reduced availability at higher set sizes. Masking does not reliably alter guess rate at any load, and no load X mask interaction was observed. Error bars denote within-subject 95% Cousineau-Morey CIs across the full 2x3 design.

4.2.2 Guess rate

A 2 X 3 ANOVA on guess rate revealed a significant main effect of load, $F(2, 56) = 3.877$, $p < .05$, $BF_{10} = 0.208$, reflecting increased guessing at higher set sizes (**Figure 4B**). Critically, there was no main effect of mask, $F(1, 28) = 2.165$, $p > .1$, $BF_{10} = 0.193$, and no load X mask interaction, $F(2, 56) = 0.385$, $p > .5$, $BF_{10} = 0.009$ (**Figure 3C**).

4.2.3 Effective precision

As in Experiment 1, an effective precision measure (see **Supplementary S2**) was computed for each participant at each load and mask condition. A 2 X 3 ANOVA revealed a significant main effect of load ($F(1.67, 46.6) = 3.48$, $p < .05$) but no main effect of mask and no load X mask interaction (both $BF_{10} < 0.18$). Collapsed across

load, effective precision did not differ between masking conditions ($t(28) = -1.615$, $p > .1$, $BF_{10} = 0.68$).

Thus, the same interference stream that selectively eliminated access to iconic representations had no reliable effect on working memory maintenance. This was true despite clear load sensitivity: increasing set size impaired performance, but masking neither scaled with load nor impaired performance at any set size.

5. Experiment 3

Experiment 3 held the encoding regime, retention interval, task, and mask properties of Experiment 1 constant, varying mask duration with offset locked to the probe so longer masks began earlier, at sample offset and shorter masks began later in the delay (**Supplementary S3**). This isolated mask onset as the sole determinant of whether interference coincides with the iconic-vulnerable window.

5.1 Experiment 3 Methods

Each trial consisted of a 50ms sample grating followed by a fixed 400ms retention interval. Within the mask condition, mask duration varied across five levels (50ms, 100ms, 200ms, 300ms, 400ms). Mask offset was anchored to probe onset such that mask onset time varied as a function of duration (onset = 400 - duration). Thus, longer durations resulted in earlier and more sustained occupation of the same feature-selective channels during retention.

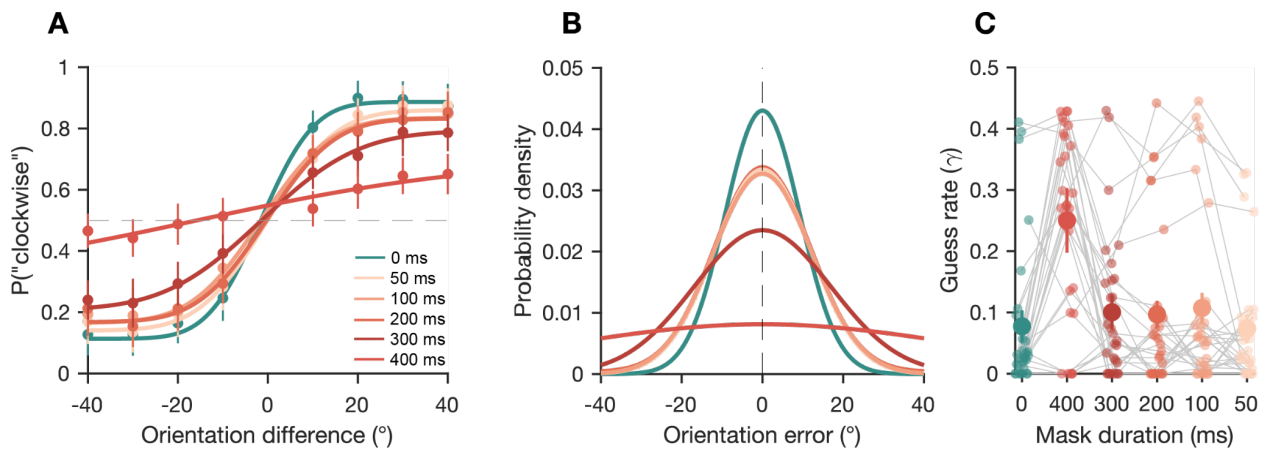


Figure 5. Masking impairment depends on mask onset. (A) Group psychometric functions for the unmasked condition (teal) and each of five mask durations (light = 50ms to dark = 400ms) in Experiment 3. Points show the mean proportion of "clockwise" responses across participants at each signed orientation difference; solid curves are cumulative Gaussian fits. Error bars denote 95% between-subject

confidence intervals. **(B)** Precision depicted as Gaussian probability density functions. Curves depict the internal noise distribution implied by fitted sigmas from group-level psychometric fits in (A). **(C)** Guess rate (γ). Individual participant data (small dots, connected across conditions by thin gray lines); group means with within-subject 95% Cousineau-Morey confidence intervals (large dots with error bars).

5.2 Experiment 3 Results

Collapsed across mask duration, precision did not differ between the mask and no-mask conditions, $t(28) = -0.72$, $p = .48$, $BF_{10} = 0.24$. In contrast, masking reliably increased guess rate, $t(28) = 3.46$, $p = .002$, $BF_{10} = 32.68$. To characterize the temporal profile, we ran a one-way repeated-measures ANOVA on guess rate across all six conditions (no-mask + five mask durations), which revealed a reliable condition effect, $F(2.55, 71.3) = 18.77$, $p < .001$. Follow-up Bonferroni-corrected pairwise comparisons against the no-mask condition revealed a difference only in the 400ms condition, $t(28) = 5.70$, $p < .001$, $BF_{10} = 13,157.66$ (all other p 's $> .1$).

Because a linear duration trend on guess rate was statistically reliable in the full dataset (slope = $.00039/\text{ms}$, $t(143) = 6.37$, $p < .001$), we ran a sensitivity analysis excluding the 400ms condition; the trend did not survive (slope = $.00007/\text{ms}$, $t(114) = 1.15$, $p > .1$, $BF_{10} = 0.18$), confirming that the effect was driven entirely by the 400ms condition.

In Experiment 3, only the 400ms mask (the single condition in which mask onset coincided with sample offset) impaired report. Masks beginning 100ms or more after sample offset, even when they occupied the same feature channels for most of the delay period, had no measurable effect on performance.

6. Discussion

The present experiments asked a direct question about representational format: does visual working memory storage depend on the same sensory-format code that supports iconic persistence? The answer, across all measures and load levels, was no.

In Experiment 1, dynamic, feature-matched masks delivered during a 400ms iconic retention interval tripled guess rates relative to unmasked trials while leaving the precision of surviving representations intact, consistent with wholesale disruption of a fragile sensory-format trace. In contrast, working memory performance showed no such signature. Identical, feature-matched masks, sustained throughout a five-second working memory delay, neither raised the probability of information loss nor altered the precision of retained representations at any memory load. A direct between-experiment comparison confirmed the dissociation (**Supplementary S4**): that is, the same masking

condition that disrupted iconic persistence left working memory intact. Experiment 3 reproduced this dissociation within a single iconic-scale paradigm: the mask condition matching Experiment 1 replicated its disruption, while mask onsets just 100ms later replicated the null effects observed in Experiment 2.

6.1 The behavioral signature of sensory-format maintenance

Experiment 1 established the signature of sensory-format vulnerability in this paradigm. Feature-matched masks reliably disrupted the availability of iconic representations, providing direct evidence that the masking manipulation engages the relevant sensory channels during maintenance itself ([Sreenivasan & Jha, 2007](#); [Bruchmann et al., 2010](#); [Rademaker et al., 2015](#)). Experiment 2 tested whether vulnerability to masking would scale with working memory demands by crossing mask condition with memory load (1–3 items). Expected load effects were observed: precision declined and guessing increased as set size increased, confirming that the paradigm was sensitive to capacity constraints. Critically, neither precision nor guessing was affected by mask presence at any load. This interaction (load sensitivity without mask sensitivity) is difficult to accommodate under any account in which working memory storage remains coupled to the same channels targeted by incoming sensory input.

6.2 The transition from sensory-format dependence is fast and sharp

Experiments 1 and 2 differ in encoding duration and retention interval; Experiment 3 held both parameters constant, varying only mask onset, and replicated the same dissociation. Within subjects and within the same 400ms delay, mask onsets timed to sample offset replicated Experiment 1's disruption, while masks onset 100ms or later had no effect. That is, a 300ms mask occupying the same feature channels and retinotopic locations for three-quarters of the retention interval produced no measurable cost, while a 400ms mask in the same paradigm eliminated usable information on a substantial number of trials. If a single, sensory-format representation were progressively degraded by ongoing interference, longer masks should produce larger costs; they did not. The window within which report depends on the interference-vulnerable substrate is therefore narrow, with a sharp, rather than graded transition. This profile implicates a handoff between representational formats, with the sensory-format trace giving way to a code that tolerates continued drive of the same channels.

6.3 A boundary condition on accounts of perceptual reuse

This dissociation constitutes direct behavioral evidence against any sensory recruitment account in which working memory maintenance depends on the continued integrity of a

stimulus-coupled perceptual code occupying the same feature channels and retinotopic positions as the remembered target (Serences, 2016; Sreenivasan et al., 2014).

While these results establish a critical boundary condition on sensory-based storage accounts, several contemporary alternatives are consistent with a representational format that tolerates ongoing feature-matched interference. For instance, the perceptual code established during encoding may be recoded into an orthogonal format that reduces representational overlap with the incoming sensory stream (Murray et al., 2017; Rademaker, Chunharas, & Serences, 2019), consistent with rotational dynamics that separate sensory from mnemonic codes within overlapping populations (Libby & Buschman, 2021). Alternatively, working memory contents may be stored within activity-silent synaptic states (Stokes, 2015; Wolff et al., 2017), or distributed frontoparietal circuits, with delay-period activity in early visual cortex reflecting feedback rather than storage (Bettencourt & Xu, 2016; Lorenc et al., 2018; Postle, 2006).

These results are also consistent with recent evidence that working memory representations in parietal cortex are maintained in a recoded, abstract format that retains task-relevant feature information decoupled from the stimulus-evoked code (Duan & Curtis, 2024). The behavioral dissociation we report is consistent with this kind of abstraction: a maintained representation no longer coupled to the same feature channels would preserve report while tolerating feature-matched interference at the remembered location.

6.4 Behavioral dependence as a complement to decoding evidence

The present findings bear on a recurrent methodological tension in this literature. Much of the evidence for sensory recruitment comes from multivariate decoding: feature-specific delay-period activity can be read out from early visual cortex, and the fidelity of that readout covaries with behavioral precision (Harrison & Tong, 2009; Serences et al., 2009; Emrich et al., 2013; Ester et al., 2013). Importantly, decodability does not establish dependency: a cortical region may carry feature information (e.g., reflecting feedback, attentional modulation, or readout preparation) without that information constituting the representational substrate that report depends on (Riggall & Postle, 2012; Xu, 2017).

Here, we provide a complementary behavioral approach by first demonstrating that a given interference stream reliably disrupts the sensory-format substrate, then asking whether a second regime shares that vulnerability. Working memory's insulation from the interference that eliminated iconic availability is evidence that maintenance does not

depend on the same retinotopically organized feature-selective channels that support iconic retention.

6.5 Conclusions

Sensory recruitment captures a genuine organizational principle of the visual system: early visual cortex provides the kind of high-dimensional, feature-selective code that is well-suited to representing perceptual content. The present results establish a boundary on how far this principle extends: once consolidation is complete, multi-second retention does not depend on the continued integrity of a sensory-format trace that shares the channels and vulnerability profile of iconic memory.

Iconic persistence and working memory maintenance are therefore distinguished not only by distinct time courses, but by the representational format and interference profile that characterize each. Any successful account of visual working memory must therefore contend with the representational discontinuity between perception and memory: the feature-selective, retinotopically-organized format that supports perception is not the code that supports memory across delay.

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